Mechanics of machines

offset crank-slider mechanism

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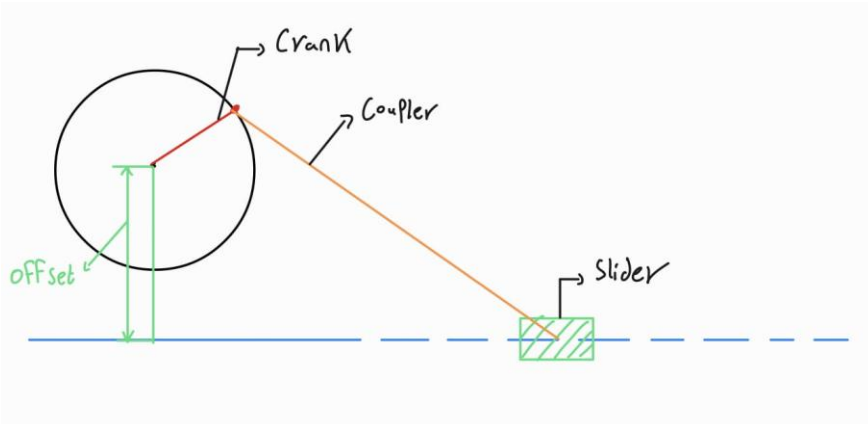
Introduction

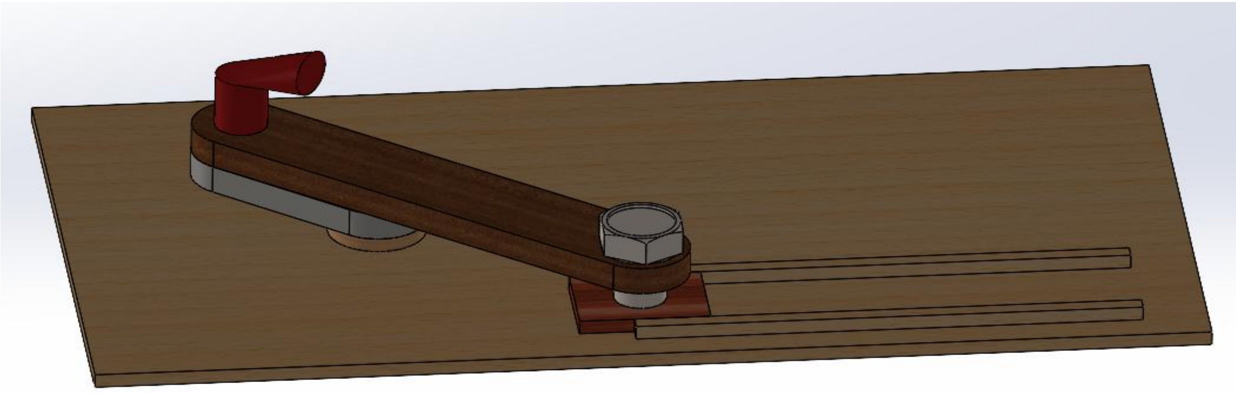
This report details the design and analysis of an offset slider-crank mechanism. This specific mechanism is a clever engineering tool used to convert circular rotation into back-and-forth linear motion. Because the slider is "offset" (positioned slightly above or below the center of the crank), it creates a unique timing profile often used in engines and industrial compressors.

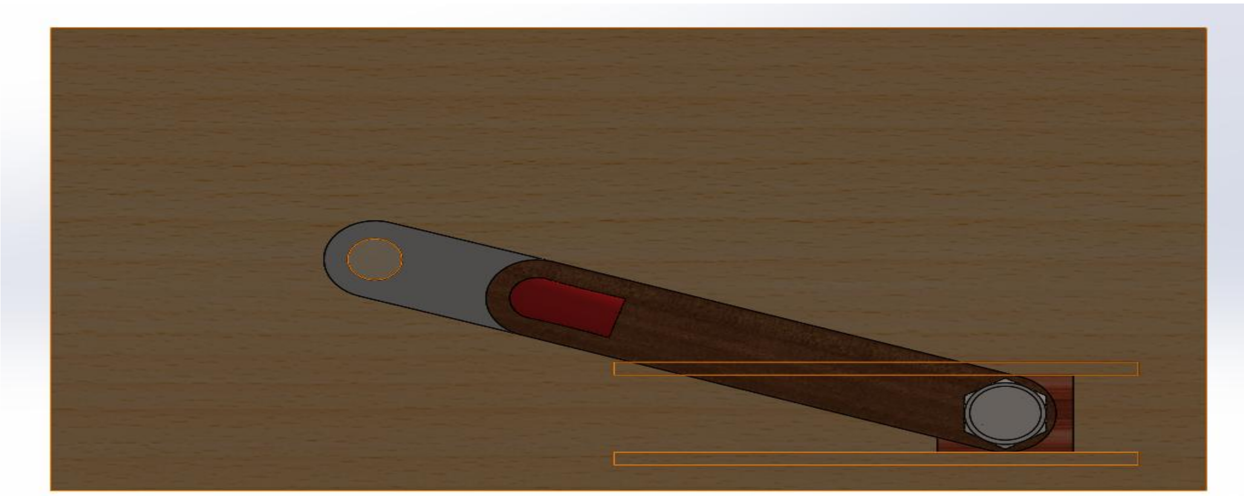
The goal of our project was to move from mathematical theory to a working physical prototype. Within this **report, we will cover:**

- **Design & Application:** Why do we choose the offset configuration and where it is commonly used in industry.
- **Kinematic Modeling**: The essential equations used to track the position, stroke length, and movement of the mechanism at every degree of rotation.
- **Calculations:** The mathematical verification of our design to ensure smooth operation.



- **Physical Results**: An evaluation of our hands-on model, comparing how it performed versus our theoretical predictions.

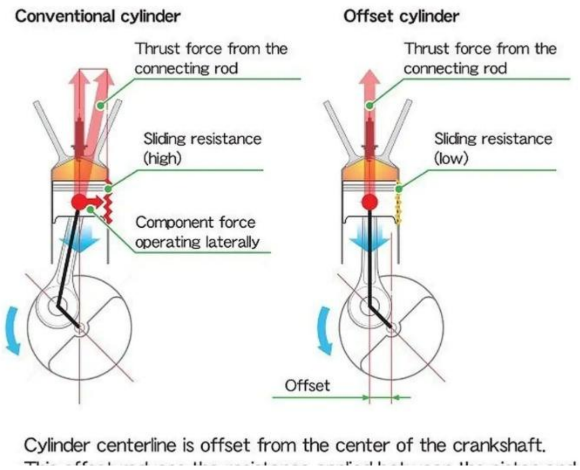




Application

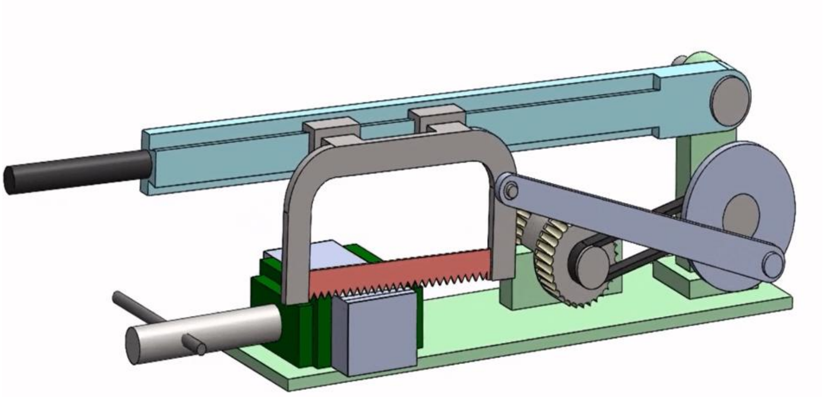
1. Internal Combustion Engines

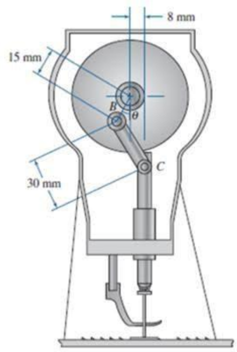
In many modern car engines, the cylinder axis is slightly offset from the crankshaft center. This design reduces the side-thrust forces acting on the cylinder walls during the high-pressure power stroke. By reducing this friction, the engine becomes more efficient, experiences less wear and tear over time and runs slightly quieter.



2. Reciprocating Saws and Power Hacksaws

If you've ever used a motorized saw, you might have noticed it seems to "bite" on the forward stroke and move quickly on the backstroke. An offset slider-crank is often used here to create a quick-return motion. The goal is to spend more time on the slow, powerful "cutting" stroke and less time on the "return" stroke where no work is being done, making the tool much more productive.





3. Sewing Machines

In both domestic and industrial sewing machines, the needle-drive assembly frequently employs an offset slider-crank mechanism. The offset deliberately alters the kinematic motion of the needle, precisely controlling the timing between the needle penetrating the fabric and the rotary hook catching the thread loop to form a stitch below. This adjusted stroke profile ensures smooth, consistent fabric penetration and allows the machine to maintain reliable thread tension even at high operating speeds.

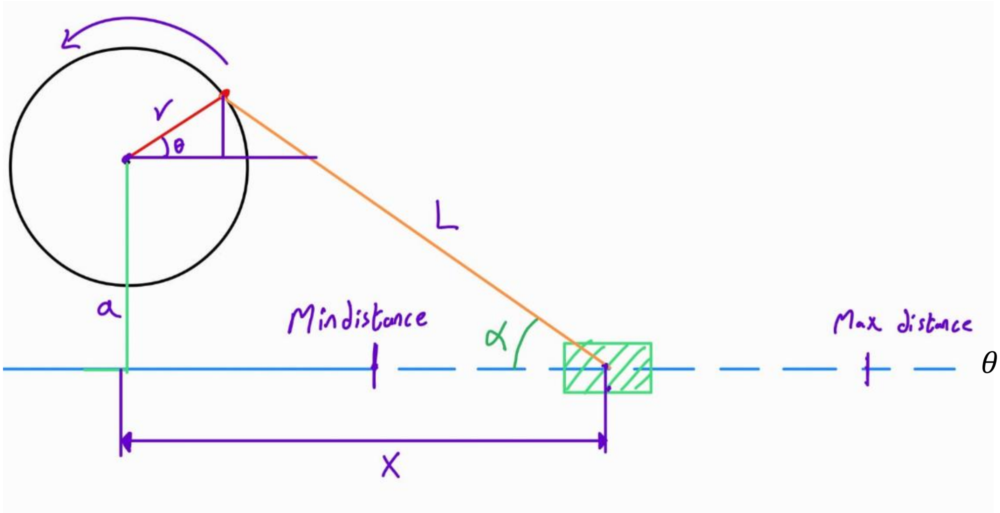
Equation

Equation 1

$$x = rcos(\theta) + L\cos(\alpha)$$

Equation 2

$$r\sin(\theta) + a = L\sin(\alpha)$$



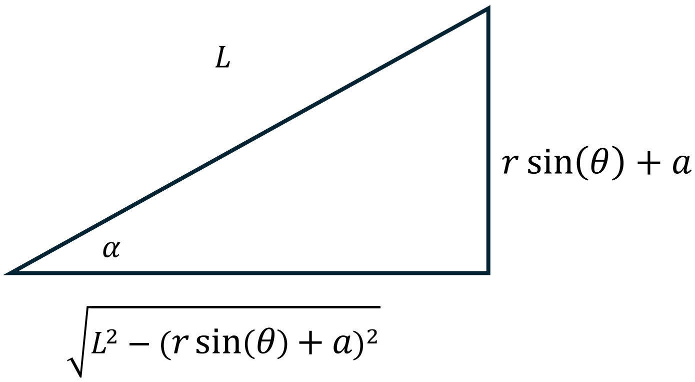
$r = length\ of\ crank$

$L = length\ of\ coupler$

$a = off\ set\ disatnce$

$\theta = angle\ of\ crank\ (C.C.W)$

$$\mathrm{L}\cos(\alpha) = \sqrt{L^2 - (r\sin(\theta) + a)^2}$$



From Eq 1&2

$$x = rcos(\theta) + \sqrt{L^2 - (r\sin(\theta) + a)^2}$$

Distance traveled by slider

$$D = x_{min} - x(\theta)$$

Calculations

We used the following dimensions to make our calculations

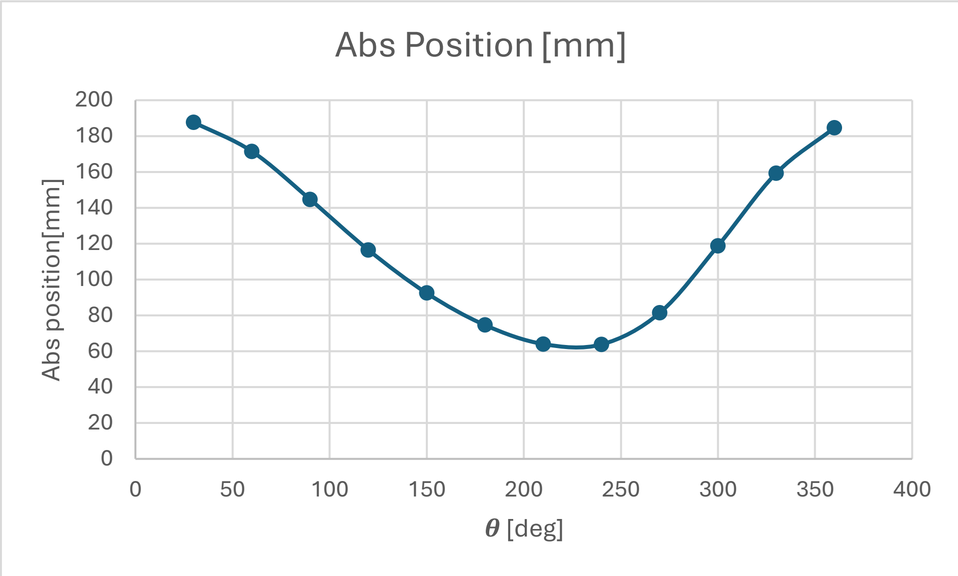
And physical model:

r $(length\ of\ crank) = 55$ mm   L$(length\ of\ coupler\ ) = 145$ mm
a$(off\ set\ disatnce) = 65$ mm

Position calculation

$$x = 55cos(\theta) + \sqrt{145^2 - (55\sin(\theta) + 65)^2}$$

Position table	
θ [deg]	Abs Position [mm]
30	187.698
60	171.456
90	144.655
120	116.456
150	92.435
180	74.615
210	64.032
240	63.819
270	81.394
300	118.819
330	159.295
360	184.615

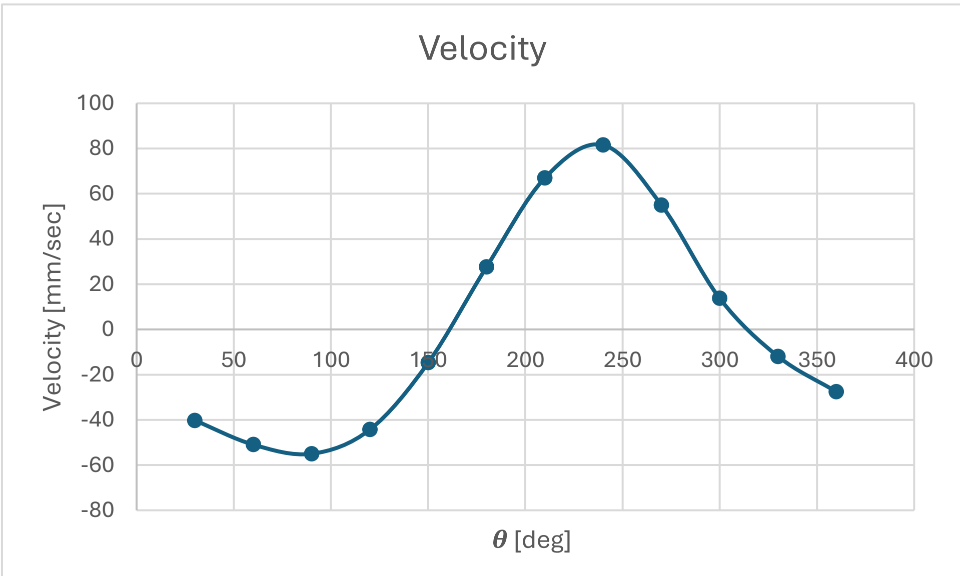


Velocity calculations

$$x = 55cos(\theta) + \sqrt{145^2 - (55\sin(\theta) + 65)^2}$$

$$\frac{dx}{dt} = v = -\omega \times [55\sin(\theta) + \frac{(55\sin(\theta) + 65) \times (55\cos(\theta))}{\sqrt{145^2 - (55\sin(\theta) + 65)^2}}]$$

Velocity Table	
θ [deg]	**Velocity [mm/sec]**
30	-40.25
60	-50.95
90	-55
120	-44.31
150	-14.75
180	27.58
210	66.96
240	81.55
270	55
300	13.71
330	-11.96
360	-27.58

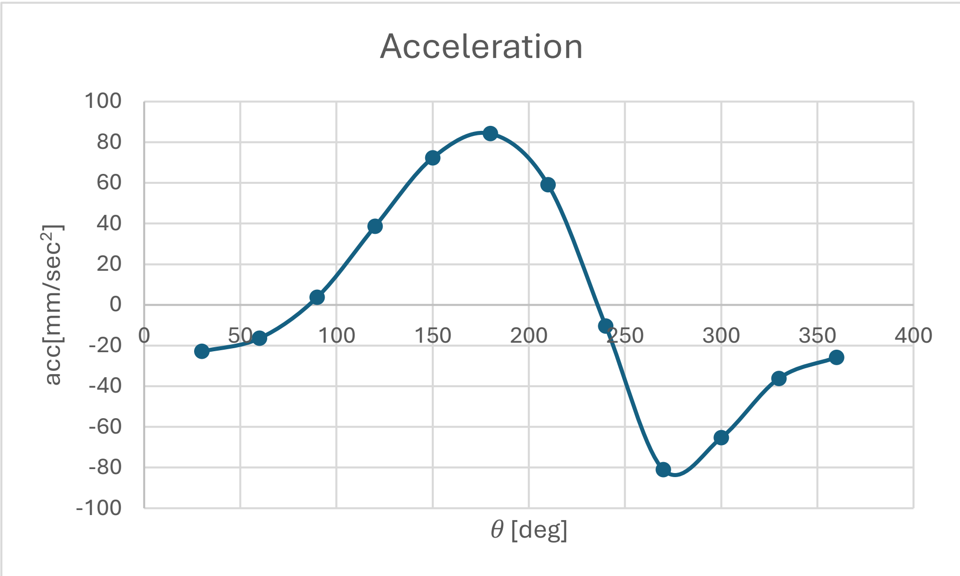


Acceleration calculations

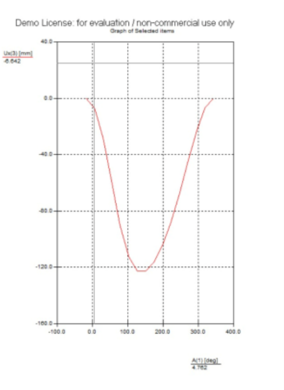
$$v = -\omega \times \left[55\sin(\theta) + \frac{(55\sin(\theta) + 65) \times (55\cos(\theta))}{\sqrt{145^2 - (55\sin(\theta) + 65)^2}}\right]$$

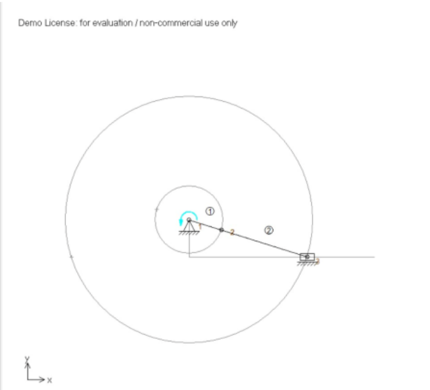
$$\frac{dv}{dt} = a = -55\omega^2 \times \left[\cos(\theta) + \frac{55\cos(2\theta) - 55\sin(\theta)}{\sqrt{145^2 - (55\sin(\theta) + 65)^2}} + \frac{(55^2\sin(\theta)\cos(\theta) + 65 \times 55\cos(\theta))^2}{(145^2 - (55\sin(\theta) + 65)^2)^{3/2}}\right]$$

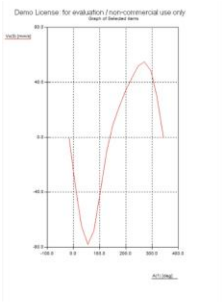
acceleration Table	
θ [deg]	acc $[mm/sec^2]$
30	-22.91
60	-16.42
90	3.8
120	38.58
150	72.35
180	84.21
210	59.11
240	-10.37
270	-81.09
300	-65.37
330	-36.15

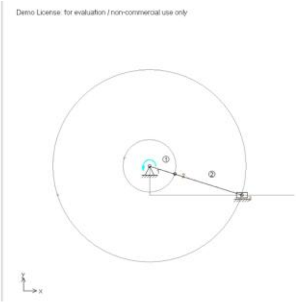


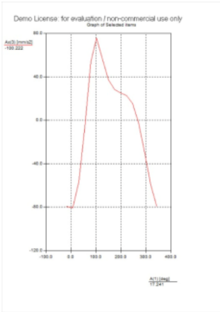
Analysis

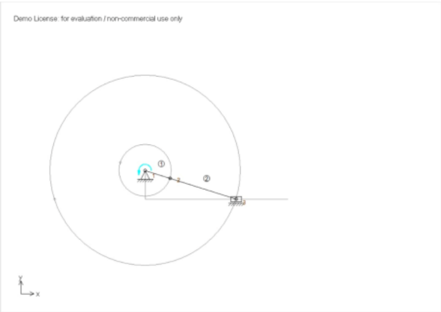


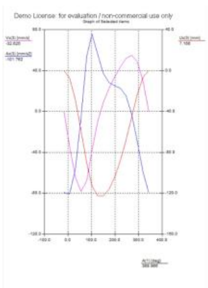


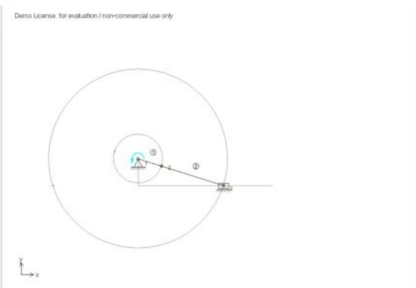








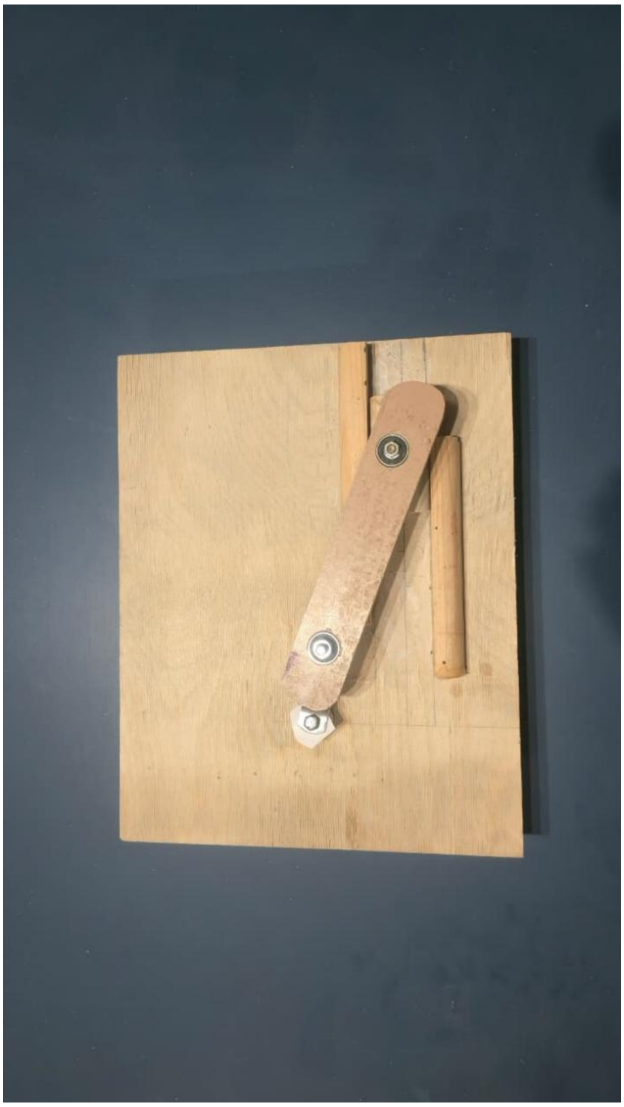


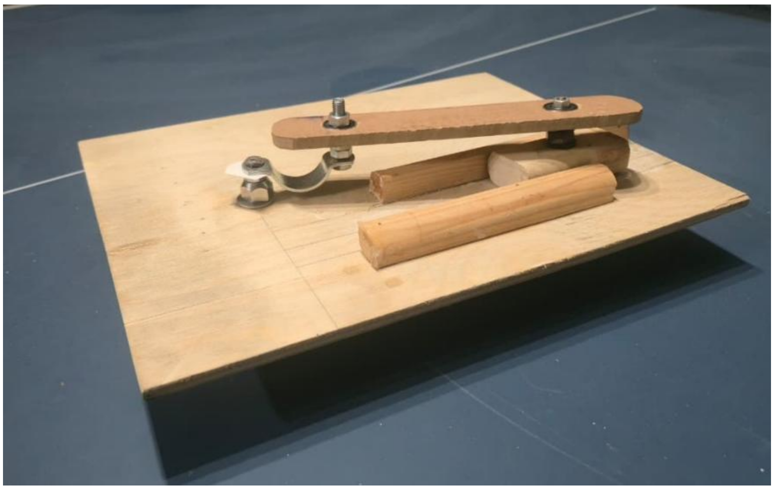


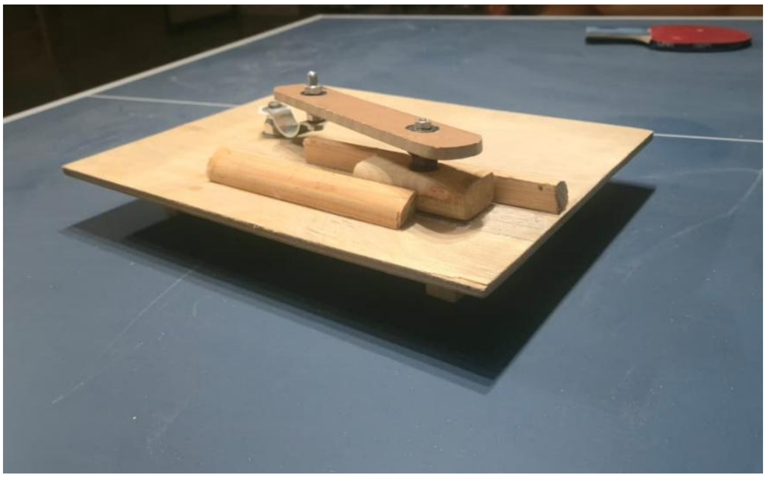
Result listing SAM 7.0.97 Mechanism: offset slider-crank

A(1)	Ux(3)	Vx(3)	Ax(3)	
[deg]	[mm]	[mm/s]	[mm/s2]	
-17.458	0.000	0.000	-79.606	
6.542	-7.174	-34.359	-80.766	
30.542	-28.137	-64.120	-56.896	
54.542	-58.673	-78.207	-6.328	
78.542	-90.136	-67.808	54.084	
102.542	-112.731		-38.620	75.723
126.542	-122.690		-10.360	56.472
150.542	-122.713		8.905	37.310
174.542	-116.040		22.359	28.393
198.542	-104.304		33.497	25.349
222.542	-88.116	43.665	22.863	
246.542	-67.996	51.925	15.253	
270.542	-45.353	55.007	-2.424	
294.542	-23.286	48.481	-29.949	
318.542	-6.499	29.633	-59.482	
342.542	0.000	-0.000	-79.606	

Physical model







Results and comments

Results

1. **Kinematic Validation:** The completed physical model successfully demonstrated the predicted motion of the offset slider-crank mechanism. The introduction of the offset (a) effectively altered the stroke timing, confirming the theoretical quick-return characteristics where the forward stroke took visibly longer than the return stroke.

2. **Stroke Length Verification:** The inner and outer dead center positions achieved by the physical slider closely matched the theoretical maximum and minimum absolute positions calculated in the position table (ranging from a minimum of approximately 74.17 mm to a maximum of 184.01 mm).

Comments and Observations:

1. **Friction and Tolerances:** While the theoretical calculations and CAD simulations assume rigid links and frictionless joints, the physical prototype experienced minor mechanical losses. Slight clearances

in the pin joints and friction along the slider's offset track may cause slight deviations from the ideal, mathematically perfectly smooth velocity and acceleration curves.

2. **Transmission Angle:** During the physical operation, it was observed that the offset affects the transmission angle (the angle between the connecting rod and the slider axis). Extreme offsets can lead to a poor transmission angle, increasing side-thrust and the risk of the slider binding in its track. Our chosen dimensions maintained a functional transmission angle throughout the full 360 rotation.

Conclusion:

Moving from mathematical theory to a physical prototype confirmed that an offset slider-crank is a highly effective method for altering the time ratio of a mechanism. The project successfully bridged the gap between kinematic equations and practical mechanical design, proving the viability of this linkage for applications requiring a slow, powerful working stroke and a rapid return, but unfortunately there is a slight deviation and error from the theoretical calculations and the actual one as the lengths may differ from the CAD/simulation and the physical model due to manual manufacturing and human error